Scientific Research Methodology
6. Research problems, objectives, and questions

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Discussion

- research and its motivation: a gap in knowledge in a field
- research problem: a situation that is not fully understood
- the problem must be **researchable** using scientific methods
- Examples
  - prove or disprove a mathematical result
  - find an innovative algorithm to solve a particular computing problem
  - explain a physical or chemical phenomenon observed in an experiment

The problem

⋆ For research to have an impact, it must address a problem that other members of the community also want to solve.

Practical problem: What should be done?

- Problem: the bicycle has broken down
- Search: repair shop, bicycle parts shop
- Search result: the address found
- Practical solution: find the part and replace it

Conceptual problem: What should we think about, and how?

The structure of problems

- a situation or condition
- unwanted consequences caused by that condition; costs we want to avoid

Practical problems:
Condition: a. I missed the bus.
b. The ozone layer is thinning.
Cost: a. I will be late for work and risk losing my job.
b. An increased risk of skin cancer.

The reader's question: So what?
Anticipating the question: provide the justification *a priori*.
Success: the reader accepts that they have this problem and wants to know the solution.

Practical vs. conceptual problems

Practical: anything that has a cost for you or your readers.
Conceptual: something is unknown or not understood.

A conceptual problem has no tangible cost. But it has a **consequence**: a form of ignorance, a gap in knowledge.

Finding new questions or problems

Problem P2 can be solved if problem P1 is solved. So let us investigate problem P1.

"Pure" vs. "applied" research

Pure research: a conceptual problem, not necessarily connected to a practical situation.

1. Topic: I am studying electromagnetic radiation in a region of the universe
2. Question: I want to find out how many galaxies are in that region
3. Significance: I help readers understand how the universe behaves.

Applied research: a conceptual problem with practical consequences

1. I am studying differences between images from Hubble and ground-based telescopes
2. Question: I want to determine how much the atmosphere distorts observations
3. Significance: I want to help astronomers use ground-based telescope data more effectively

How do you find a good research problem?

- ask more experienced researchers for help
- look for problems in your sources
- examine the conclusions of your studies: related problems can often be investigated

What if you lack experience?

- uncertainty and anxiety are normal! They reflect inexperience, not incompetence
- write while you read and research: this makes the final writing easier
- break the task into smaller steps
- pursue realistic objectives!
- accept that the whole process is a learning experience

From problems to sources

Primary sources: original documents, the central object of the research
Secondary sources: books, articles, and documents based on primary sources; the literature of the field
Tertiary sources: overviews, articles, and books based on secondary sources. Generally aimed at broad audiences, simplified, and providing a basic overview (e.g. Wikipedia)

Evaluating sources

- Relevance: will I find useful information?
  - keywords, introduction, abstract
- Reliable or not? Published online or by a prestigious publisher? Peer-reviewed? Author's reputation? Only available online? Recent? Does it include references?
- is the source cited by others?

Taken from "The Craft of Research", Booth et al.

Error is bad, but dishonesty is worse. One of Booth's students got a summer job with a drug company and was assigned to go through stacks of doctors' answers to questionnaires and shred certain ones until nine out of ten of those left endorsed the company's product. These bogus data were then used to "prove" that the product worked. The student quit in disgust and was, no doubt, replaced by someone less ethical.

Working with sources you read and use

- Store bibliographic information: author, title, volume, year, etc.
- Create reusable databases: BibTeX and other options
- Extend results from a source, find arguments related to the problem, and look for data
- decide whether to quote directly, paraphrase, or mention a source

Research objectives

- what the research is intended to achieve
- define objectives using action verbs:
  - determine
  - analyse
  - verify
  - compare
- general (overall progress of the study), specific (concrete, measurable steps)

Research questions

- State what you want to find out
- concise, testable, focused, and feasible to investigate

Good questions

- have a clear answer (are not philosophical)
- **open** (answers in one's own words, complex perspectives, in-depth exploration) vs. **closed** (predefined options, yes/no, multiple choice)
- can be investigated with the available resources
- are neither too general nor too trivial

Types of research question

1. **Descriptive:** What is happening . . . ?
2. **Comparative:** What are the differences between . . . ?
3. **Explanatory:** What influences . . . ?
4. **Predictive:** To what extent can we predict . . . ?
5. **Evaluative:** How effective is . . . ?

The formulation process

1. Identify the problem
2. Set the general objective
3. Break it down into specific objectives
4. Formulate research questions for each objective

Common errors

- too many objectives, or objectives that are too vague
- questions that cannot be investigated (resources are unavailable)
- confusing objectives with tasks (work activities)
- tautological formulation (the objective is to prove what we assumed at the outset)

Application

The problem

Ninth-grade pupils have difficulty understanding linear functions, and their mathematics test results are declining. Traditional teaching methods are suspected of being insufficiently interactive.

General objective: To evaluate how effectively an interactive teaching method improves ninth-grade pupils' understanding of linear functions.

Specific objectives

O1: To identify the initial level of understanding of linear functions.
O2: To implement an interactive teaching method using visual activities and dynamic software.
O3: To compare pupils' performance before and after the intervention.
O4: To analyse pupils' perceptions of the new method.

Research questions

Q1 (for O1): What is the pupils' level of proficiency in linear functions before the intervention?
Q2 (for O2): How is the interactive method implemented, and which components are perceived as useful?
Q3 (for O3): Are there significant differences between test results before and after the interactive activities?
Q4 (for O4): How do pupils perceive the effectiveness and appeal of the interactive teaching method?